

Ali Haydar Sucu

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📍 Ankara, Turkey

Education

Ankara Yıldırım Beyazıt University

Bachelor of Science in Computer Engineering

Ankara, Turkey

Sep 2023 – Jun 2028 (Expected)

- Current GPA: 3.13/4.00
- Activities: AYBU Hezarfen Student Club, AYBU Turkish World Youth Club

Tevfik İleri Anatolian High School

High School Diploma (Graduated with 96.43/100)

Ankara, Turkey

Sep 2019 – Jun 2023

Professional Experience

AI Developer | Part-Time Student

AYBU ÖĞMER

Jun 2025 – Present

Ankara, Turkey

- Led development of an intelligent document-interactive AI assistant program (with RAG) using Llama 4 LLM with web interface published for university-wide access
- Implemented core features: document processing (PDF, DOCX, TXT), question answering, text summarization, and Turkish speech-to-text/text-to-speech capabilities
- Designed system architecture with real-time audio streaming and document embedding for knowledge retrieval

Embedded Software Developer

PUSULA USV

Oct 2024 – Aug 2025

Ankara, Turkey

- Developed and optimized autonomous navigation and control algorithms for embedded systems on NVIDIA Jetson platform using Python.
- Engineered systemd services and real-time system integration for launching multi-threaded ROS 2 nodes handling sensor data streams (Lidar, ZED Camera) and autonomous control.
- Integrated and debugged multiple embedded sensors: A2M12 Lidar sensor fusion, ZED 2 stereo camera processing, Pixhawk flight controller motor commands, and GPS.
- Collaborated with Electronics team on hardware-software co-design, conducting extensive real-world testing, debugging sensor communication protocols, and optimizing system performance

Avionics & Embedded Systems Team Member

AYBU HEZARFEN Club | Gökçen Rocket Project

Sep 2023 – Jun 2024

Ankara, Turkey

- Conducted detailed sensor research and selection for rocket avionics systems.
- Contributed to embedded flight control board design and development.
- Performed embedded system integration and testing of avionics hardware components, serial communication debugging, and sensor calibration
- Developed system specifications documentation and conducted initial flight avionics validation for Teknofest's rocket competition

Featured Projects

FocusPath - Focus Tracking & Analysis | C++, Qt 5/6, X11 APIs, SVG Rendering, Linux

2025

- GitHub: github.com/alihaydarsucu/FocusPath
- Developed low-level system monitoring application capturing X11 window events at 200ms intervals for focus tracking and distraction analysis on Linux desktop
- Engineered real-time window tracking system using direct X11 APIs, JSONL event logging, and advanced system integration for behavioral analytics
- Implemented SVG chart generation pipeline rendering usage statistics, timeline visualizations, and attention metrics for productivity analysis

STM32F103RB Embedded Systems <i>C/C++, STM32CubeIDE, HAL, ARM Architecture</i>	2024 – 2025
• GitHub: github.com/alihaydarsucu/STM32-Learning • Embedded systems coursework on NUCLEO-F103RB MCU platform covering peripherals: GPIO configuration, UART serial communication (115200 baud), PWM generation, timer operations, and HAL library utilization • Developed low-level register-level programming skills and STM32 microcontroller architecture understanding through hands-on firmware development	
Pusula Ground Control Station <i>Python, FastAPI, Real-Time Telemetry, System Integration</i>	Jul – Aug 2025
• GitHub: github.com/abdullah-aksoy/Pusula-GCS • Designed and developed embedded-friendly GCS software for autonomous USV platform with real-time telemetry processing at high update rates • Implemented command-and-control interface for embedded system operations including ARM/DISARM commands, mode switching, algorithm control, and real-time vehicle state monitoring • Integrated mission planning with geographic data processing and live tracking visualization of autonomous vehicle operations	
Personal Portfolio With CV Generation <i>LaTeX, GitHub Actions, CI/CD, System Automation</i>	2024
• GitHub: github.com/alihaydarsucu/alihaydarsucu.github.io • Engineered automated CV generation pipeline using LaTeX and GitHub Actions for continuous ATS-compatible document generation • Designed system automation workflow demonstrating CI/CD understanding and proficiency with build automation tools	

Technical Skills

Languages: C/C++, Python, Bash/Shell, Java, JavaScript
Embedded Systems: STM32, ARM Architecture, ROS 2, Jetson, Sensor Integration
Tools: Git, Linux, Docker, STM32CubeIDE, ST-LINK Debugger
Languages: Turkish (Native), English (Professional)

Certifications & Achievements

Teknofest Success Certificate <i>Teknofest</i>	Nov 2025
• Ranked 10th among 284 teams in Mavi Vatan competition	
Career Module Certificate <i>MSMB - Mimar Sinan Mühendisler Birliği</i>	July 2025
• Professional career module certification	
Data Science Bootcamp Certification <i>Google Developer Students Community</i>	Dec 2023
• Completed intensive bootcamp covering Python programming and data processing with Pandas, NumPy, and Matplotlib	

Academic Presentations

• Colak, A., Karaaslan, H., Sen, M., et al. (including Sucu, A. H.). <i>Bootcamps: Language as a Tool to Develop Global Competencies</i>. Paper presented at the 6th FLE Joint Conference Unity in Diversity, THK University, Ankara, Türkiye, 30 May 2025.	
• Karaaslan, H., Colak, A., Durna, S., et al. (including Sucu, A. H.). <i>Language as a Tool: A Challenge-Based Summer Bootcamp Model with Scrum Scaffolding to Develop Global Competencies</i>. Poster presented at the International Challenge-Based Learning Conference, Eindhoven University of Technology, Eindhoven, Netherlands, 7–8 Apr 2025. Slides	
• Colak, A., Karaaslan, H., Korkmaz, I., et al. (including Sucu, A. H.). <i>Fostering Autonomy through CLIL-Infused Bootcamps and Scrum Scaffolding</i>. Pre-recorded presentation at the 13th Lab Session on CLIL and Learner Autonomy, Research Institute for Learner Autonomy Education, Kanda University of International Studies, Chiba, Japan, 29 Nov 2024. Video	

- Karaaslan, H., Colak, A., Korkmaz, I., et al. (including Sucu, A. H.). *Language as a Tool: A Summer Bootcamp Model to Build Global Competencies*. Paper presented at the 6th International English for Specific Purposes Conference, Middle East Technical University, Turkiye, 2 Nov 2024.
- Kurt, A., Karaaslan, H., Demir, M. H., et al. (including Sucu, A. H.). *Sharpen Your Arguments: Fallacy Fighters Workshop*. Paper presented at the 5th FLE Joint Conference Unity in Diversity, Ankara Yildirim Beyazit University, Ankara, Turkiye, 7 Jun 2024. Slides
- Colak, A., Karaaslan, H., Kose, E., et al. (including Sucu, A. H.). *Beyond Self: E-Portfolios and Ecological L2 Motivation*. Paper presented at the 5th FLE Joint Conference Unity in Diversity, Ankara Yildirim Beyazit University, Ankara, Turkiye, 7 Jun 2024.